

Pizza queue

- A Pizza stand receives on average 3 orders per 10-minute interval at a sports event.
- As the owner, you are interested in understanding certain probabilities and would like to make use of Queuing Theory.
- 1) What assumption must you make regarding the number of orders received within any time window?

Pizza queue

- A Pizza stand receives on average 3 orders per 10-minute interval at a sports event.
- As the owner, you are interested in understanding certain probabilities and would like to make use of Queuing Theory.
- 1) What assumption must you make regarding the number of orders received within any time window? **The number of orders follows a Poisson distribution.**

Pizza queue

- Provided the inter arrival times follows an exponential distribution, attempt the following questions

$$\text{*remember } P(N = n) = \frac{(e^{-\lambda})\lambda^n}{n!}$$

- What is the probability that:
 - 0 pizza will be ordered in 10 minutes?
 - 1 pizza will be ordered in 10 minutes?
 - 2 pizza will be ordered in 10 minutes?
 - 3 pizza will be ordered in 10 minutes?
 - 6 pizza will be ordered in 20 minutes?
 - 9 pizza will be ordered in 30 minutes?

Pizza queue

- Provided the inter arrival times follows an exponential distribution, attempt the following questions

*remember $P(N = n) = \frac{(e^{-\lambda})\lambda^n}{n!}$

- Note you can ‘customise’ your λ or write out in terms of factor of “t”: $P(N_t = n) = \frac{e^{-\lambda t}(\lambda t)^n}{n!}$

Pizza queue

- What is the probability that:

- 0 pizza will be ordered in 10 minutes? $P(N_t = 0) = e^{-3}(3)^0/0! = 0.0498$
- 1 pizza will be ordered in 10 minutes? $P(N_t = 1) = e^{-3}(3)^1/1! = 0.1494$
- 2 pizza will be ordered in 10 minutes? $P(N_t = 2) = e^{-3}(3)^2/2! = 0.2240$
- 3 pizza will be ordered in 10 minutes? $P(N_t = 3) = e^{-3}(3)^3/3! = 0.2240$
- 6 pizza will be ordered in 20 minutes?
- 9 pizza will be ordered in 30 minutes?

Pizza queue

- What is the probability that:
 - 6 pizza will be ordered in 20 minutes?
 - 9 pizza will be ordered in 30 minutes?

*Note you can either 'customise' your λ and determine a new one for each question or write it out in terms of factor of "t": $P(N_t = n) = e^{-\lambda t} (\lambda t)^n / n!$

Pizza queue

- What is the probability that:

- 0 pizza will be ordered in 10 minutes? $P(N_t = 0) = e^{-3}(3)^0 / 0! = 0.0498$
- 1 pizza will be ordered in 10 minutes? $P(N_t = 1) = e^{-3}(3)^1 / 1! = 0.1494$
- 2 pizza will be ordered in 10 minutes? $P(N_t = 2) = e^{-3}(3)^2 / 2! = 0.2240$
- 3 pizza will be ordered in 10 minutes? $P(N_t = 3) = e^{-3}(3)^3 / 3! = 0.2240$
- 6 pizza will be ordered in 20 minutes? $P(N_t = 6) = e^{-3 \times 2}(3 \times 2)^6 / 6! = 0.1606$
- 9 pizza will be ordered in 30 minutes? $P(N_t = 9) = e^{-3 \times 3}(3 \times 3)^9 / 9! = 0.1318$

Pizza queue

- What is the probability that:
 - Up to three people will arrive within 10 minutes?
 - Between 4 and 6 people will arrive in the next 20 minutes?
- What is the probability that:
 - Between 8 and 15 minutes will pass between any two orders?

*Remember inter arrival times follow an exponential distribution but the number of arrivals in an interval is a Poisson process.

Pizza queue

- What is the probability that:

- Up to three people will arrive within 10 minutes? For $\lambda = 3$:

$$\sum_{i=0..3} P(N_t = i) = 0.6472$$

- Between 4 and 6 people will arrive in the next 20 minutes? For $\lambda = 6$:

$$\sum_{i=4,5,6} P(N_t = i) = 0.4551$$

- What is the probability that:

- Between 8 and 15 minutes will pass between any two orders? For $\lambda = \frac{3}{10}$:

$$\int_{t=8}^{t=15} \lambda e^{-\lambda t} = 0.0796$$

Tiger and Wheels – service times

- A Tiger and Wheels station can perform wheel alignment for your motor vehicle at an average rate of 20 minutes. A customer, who requires wheel alignment to be performed at the station, typically arrives every 30 minutes during normal business hours. Now calculate the following:
 - λ and μ
 - The average number of customers present in the system, call it “ L ”.
 - The average number of customers in the queue, call it “ L_q ”.
 - The average time a customer spends in the system, call it “ W ”.
 - The average time a customer spends waiting in the queue, call it “ W_q ”.
 - The percentage of time that a station is busy.
 - The probability that 0 cars are in the system, call it “ P_0 ”.

Tiger and Wheels – service times

- A Tiger and Wheels station can perform wheel alignment for your motor vehicle at an average rate of 20 minutes. A customer, who requires wheel alignment to be performed at the station, typically arrives every 30 minutes during normal business hours. Now calculate the following:
 - λ and μ
 - $\lambda = 2$ cars arriving per hour.
 - $\mu = 3$ cars serviced per hour.

Tiger and Wheels – service times

- A Tiger and Wheels station can perform wheel alignment for your motor vehicle at an average rate of 20 minutes. A customer, who requires wheel alignment to be performed at the station, typically arrives every 30 minutes during normal business hours. Now calculate the following:
 - The average number of customers present in the system, call it “ L ”. (Hint, use steady state service)

Tiger and Wheels – service times (M/M/1/GD/∞/ ∞)

- The average number of customers present in the system, call it “ L ”

$$L = \frac{\rho}{(1-\rho)} = \mathbf{2 \text{ cars on average, where } \rho = \frac{\lambda}{\mu} = \frac{2}{3}}$$

- The average number of customers in the queue, call it “ L_q ”.

$$L_q = \frac{\rho^2}{(1-\rho)} = \mathbf{4/3 \text{ cars waiting in line on average.}}$$

- The average time a customer spends in the system, call it “ W ”.

$$W = \frac{L}{\lambda} = \frac{1}{(1-\rho)\mu} = \mathbf{1 \text{ hour the average car spends in the system.}}$$

- The average time a customer spends waiting in the queue, call it “ W_q ”.

$$W_q = \frac{\rho}{\mu(1-\rho)} = \mathbf{2/3 \text{ hours average waiting time per car.}}$$

Tiger and Wheels – service times (M/M/1/GD/ ∞ / ∞)

- The percentage of time that a station is busy.

$$\rho = \frac{\lambda}{\mu} = 2/3$$

- The probability that 0 cars are in the system, call it “ P_0 ”.

$$P_0 = 1 - \frac{\lambda}{\mu} = 1/3$$

Tiger and Wheels – service times

- Management wants to investigate the financial feasibility of opening a second station with a second technician. This station will be equal in terms of quality of work, as well as the time it takes to perform it. Now calculate the following:
 - λ and μ
 - The average number of customers present in the system.
 - The average number of customers in the queue.
 - The average time a customer spends in the system.
 - The average time a customer spends waiting in the queue.
 - The percentage of time that a station is busy.
 - The probability that 0 cars are in the system.